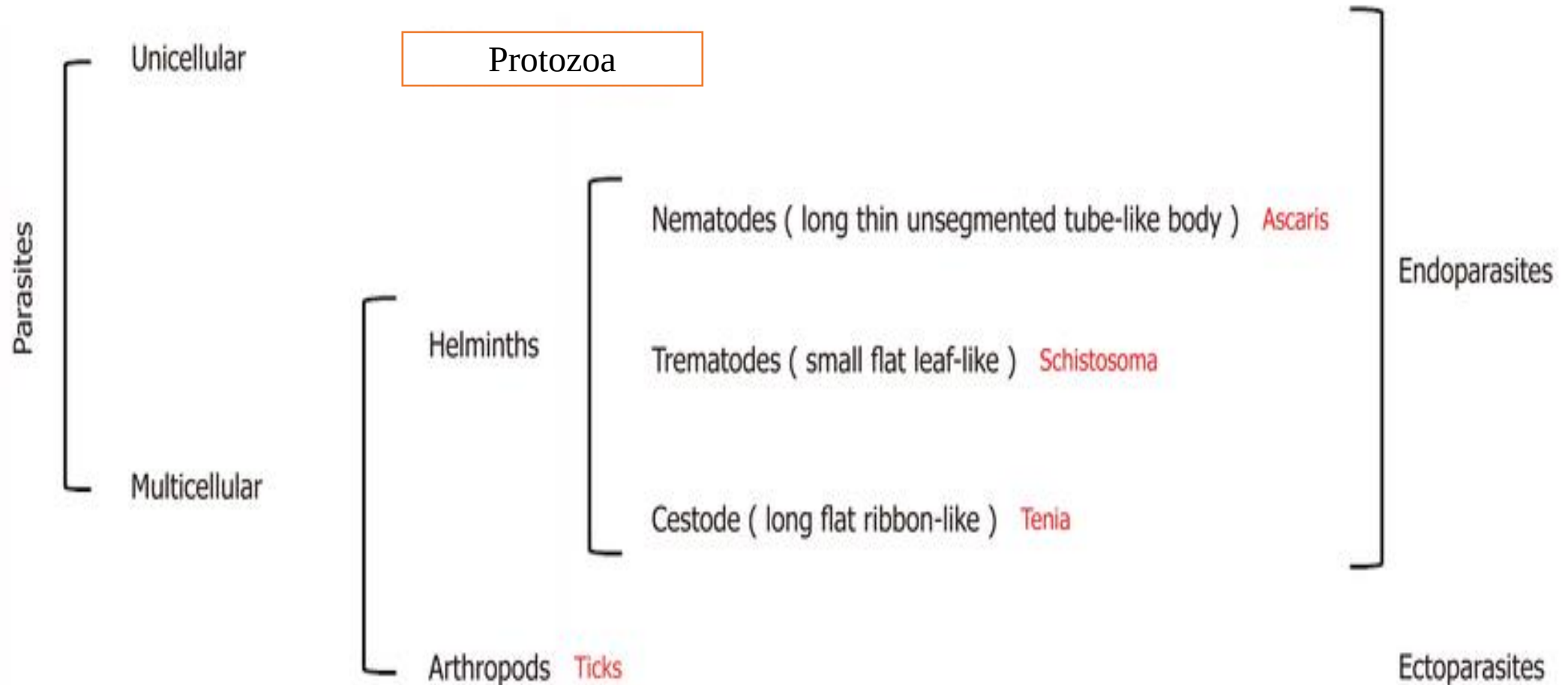




College of Medicine

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Classification of the parasites



Medical Helminthology Classification •

I- **Phylum: Platyhelminthes** (Flat worms).

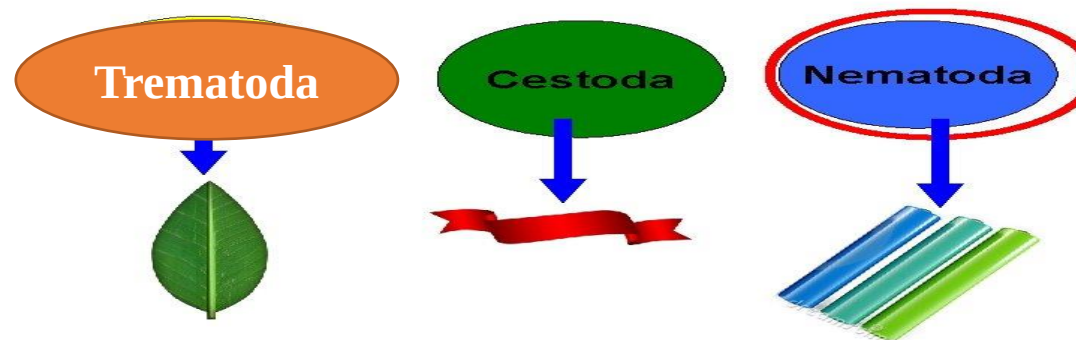
- 1) Class : **Trematoda** (Flukes).
- 2) Class : **Cestoda** (Tape worms).



II- **Phylum: Nematelminthes** (Round worms). Class: Nematoda



Medical Helminthology



General Characteristics of Helminths:

- The helminthic parasites are multicellular (metazoa), bilaterally symmetrical animals.
- The term 'helminth' (Greek helmins-'worm') originally referred to intestinal worms, but now comprises many other worms, including tissue parasites as well as many free-living species.
- Adult worms are either dioecious or monoecious.
- They vary from few millimeter to several meters.
- All have more or less developed organ system, such as alimentary, excretory and reproductive systems.

- **Trematodes** are unsegmented helminths, which are flat and broad, resembling the leaf of a tree or a flatfish (hence the name Fluke, from the Anglo-saxon word 'floc' meaning ('flatfish')).
- The name Trematode comes from their having large prominent suckers with a hole in the middle (Greek trema: hole, eidos: appearance).

Classification of trematodes based on Habitat

Blood flukes *Schistosoma mansoni*

S. hematobium

S. japonicum

Intestinal fluke *Fasciolopsis buski*

Heterophyes heterophyes

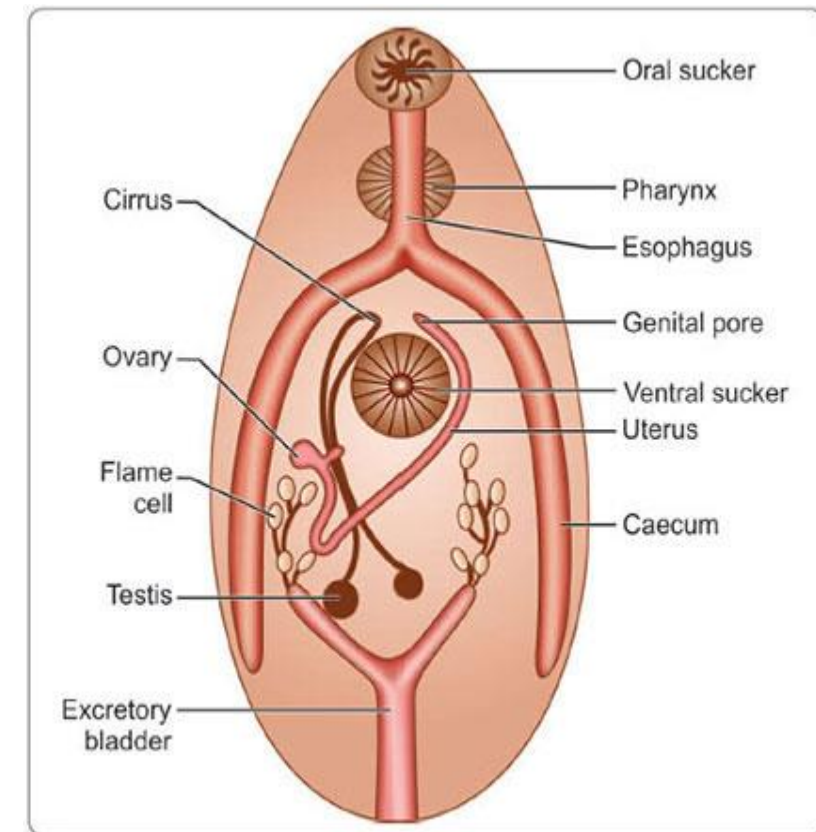
Liver fluke *Fasciola hepatica*

Clonorchis sinensis

Lung fluke *Paragonimus westermani*

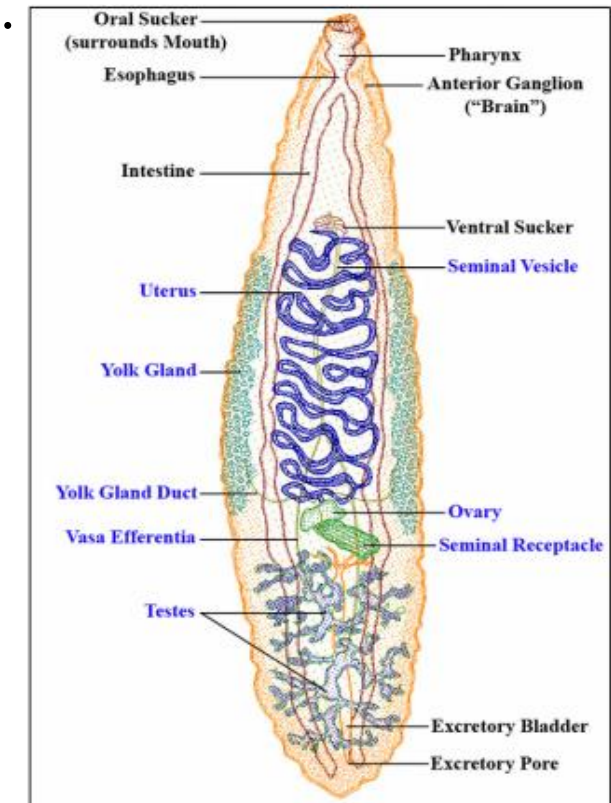
Flukes: General Characteristics

- Flat, unsegmented, leaflike
- A distinctive feature of flukes is the presence of 2 muscular cup-shaped suckers (**oral and ventral suckers** called distomata).
- The excretory system consists of flame cells and collecting tubules, which lead to a median bladder opening posteriorly.
- Size- 1mm to several centimeters.
- The reproductive system is **well-developed**



Morphology of a hermaphroditic trematode

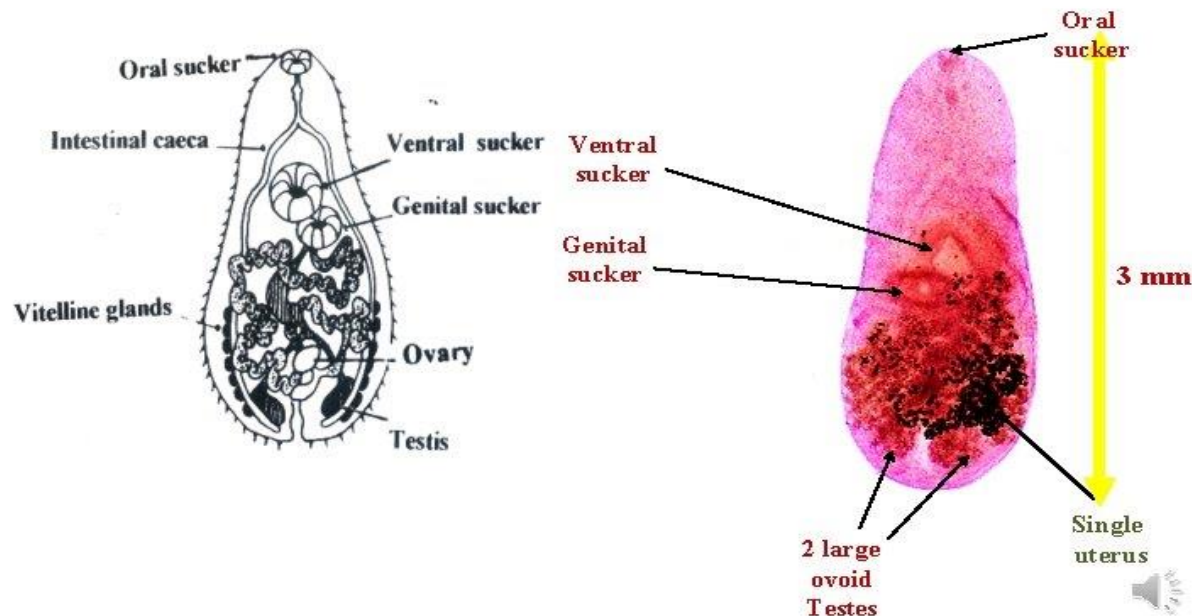
- The body is covered by an integument which often bears spines.
- Nervous system present.
- They have no body cavity, circulatory and respiratory organs.
- The alimentary canal present but incomplete: inverted Y.
- The anus is absent
- Flukes are hermaphroditic (monoecious) except for *schistosomes*, in which the sexes are separate (dioecious).



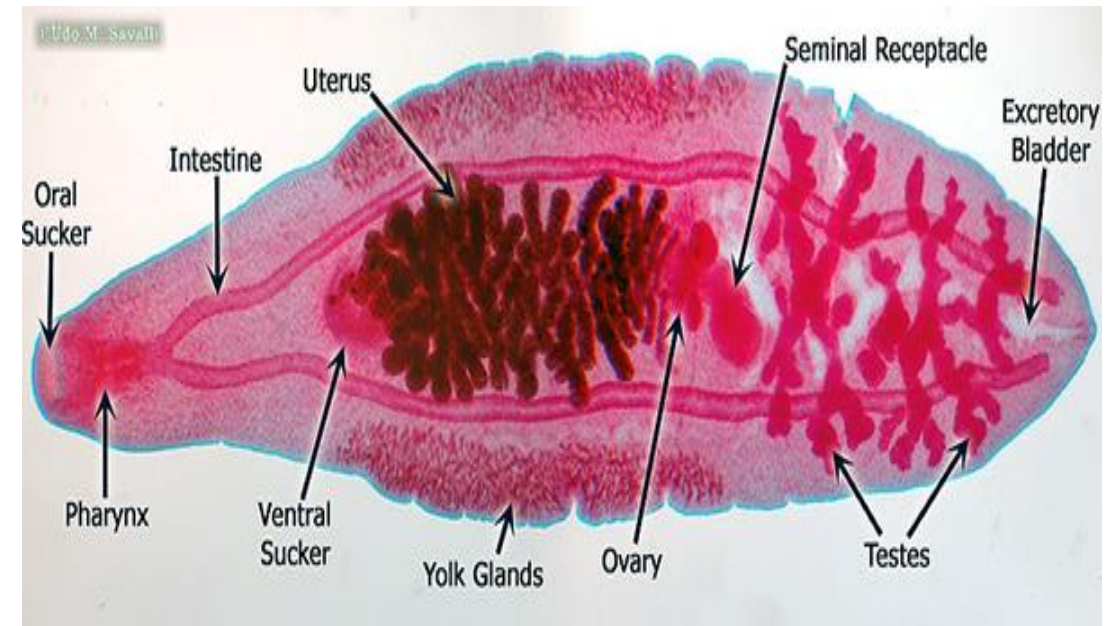
- they are digenetic require 2 hosts.
- They complete their life cycle in
- 1 definitive host (man) and 2 intermediate hosts.
- Fresh water snail act as first intermediate host and fish or crab act as second intermediate host, except schistosomes which require 2 hosts – 1 definitive host (man) and 1 intermediate host (snail).

- They vary in sizes (*Heterophyes heterophyes*), adult worm is the smallest trematodes (2 mm in length).
- Large worm size is 3cm (*Fasciola*), 5cm (*Fasiolopsis buski*)

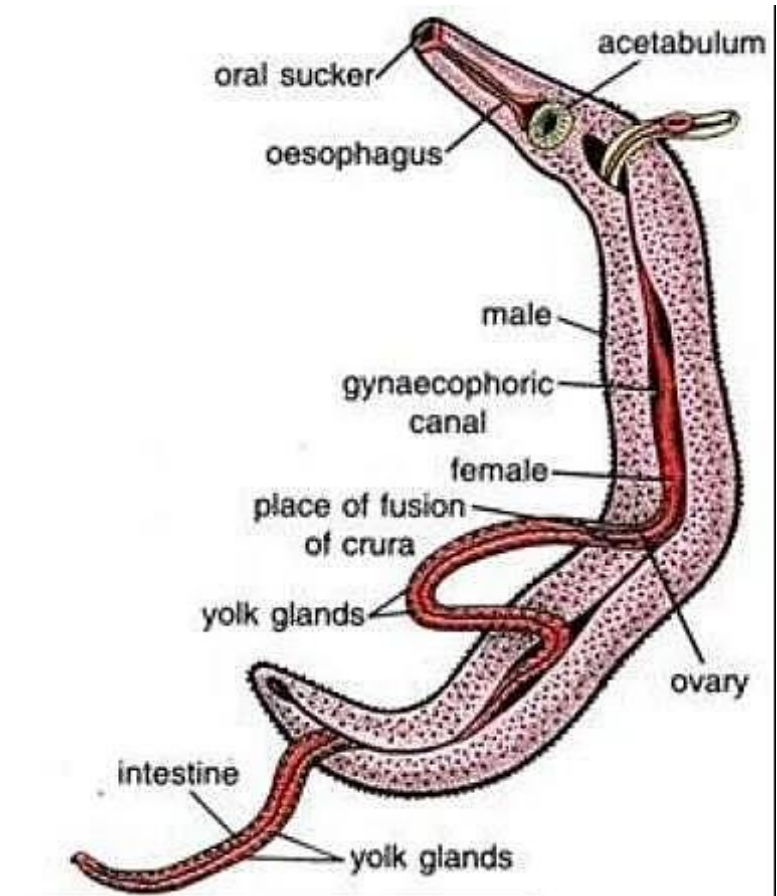
Heterophyes heterophyes



Faciola hepatica 3cm



- Some species have gynecophoral canal, male hold the female in this canal (Schistosoma).



Schistosoma. japonicum

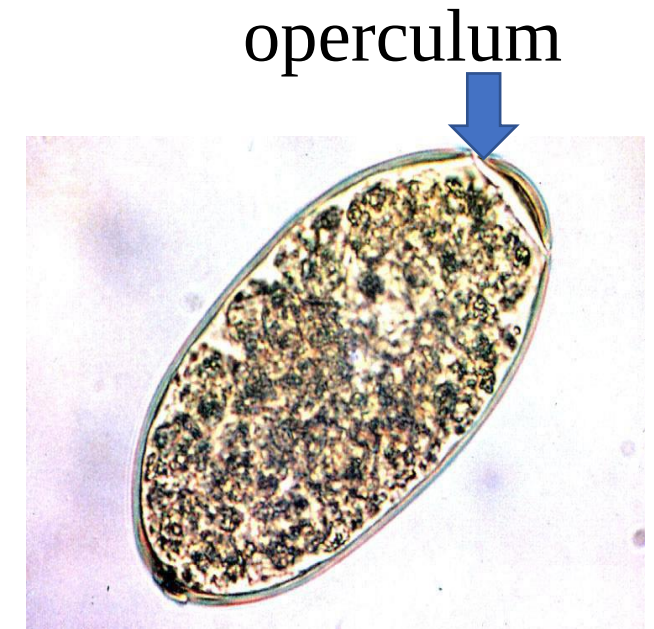
- Trematodes are oviparous and lay eggs, which are operculated, except in the case of **schistosomes** which are spine.



schistosoma haematobium
Terminal spine

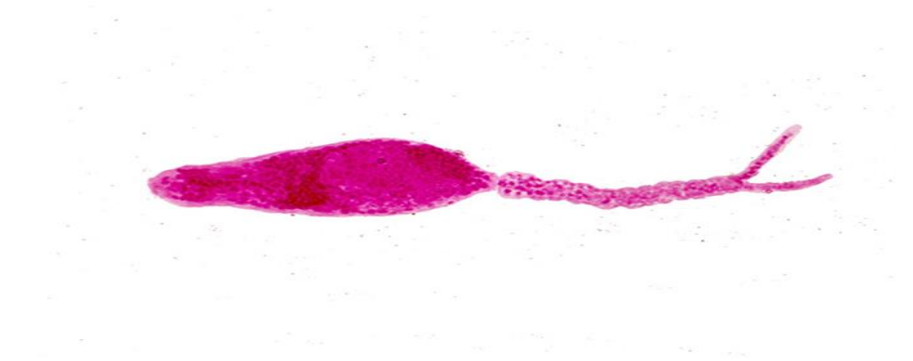


schistosoma mansoni
lateral spine



Fasciola

- In *Schistosoma* the infective stage for man and other final hosts are cercariae by penetration through the unbroken skin.



Schistosoma japonica cercaria, the stage that infects humans.

- In *Fasciola* the infective form for man and other final hosts is metacercaria. It is found on the surface of aquatic vegetations.



Liver Fluke cercaria.

BLOOD FLUKES

- **Schistosomes** are dioecious trematodes (sexes are separate), which lead to Schistosomiasis (bilharziasis).
- Clinical features: Swimmer's itch, Katayama fever, hematuria, and portal hypertension.
- Diagnosis: Detection of eggs in urine or stool, biopsy, imaging.

Schistosoma japonicum

Habitat: venules of the small intestine

Disease: oriental schistosomiasis

Egg: round with rudimentary spine laterally

Diagnostic stage



Intermediate host: are fresh water snail **Oncomelania**

Schistosoma mansoni

Habitat: in venules of the large intestine

Disease: intestinal schistosomiasis

Egg: Ellipsoidal with **lateral** spine

Diagnostic stage



Intermediate hosts: are fresh water snail

Biomphalaria

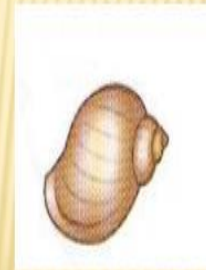
Schistosoma haematobium

Habitat: venules of the urinary bladder

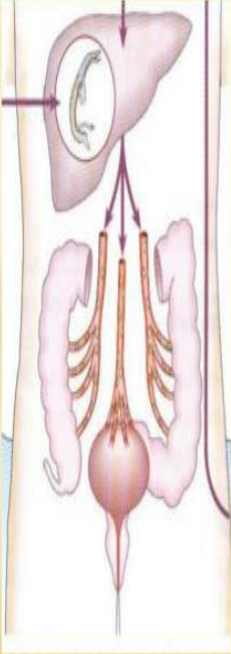
Disease: Urinary schistosomiasis

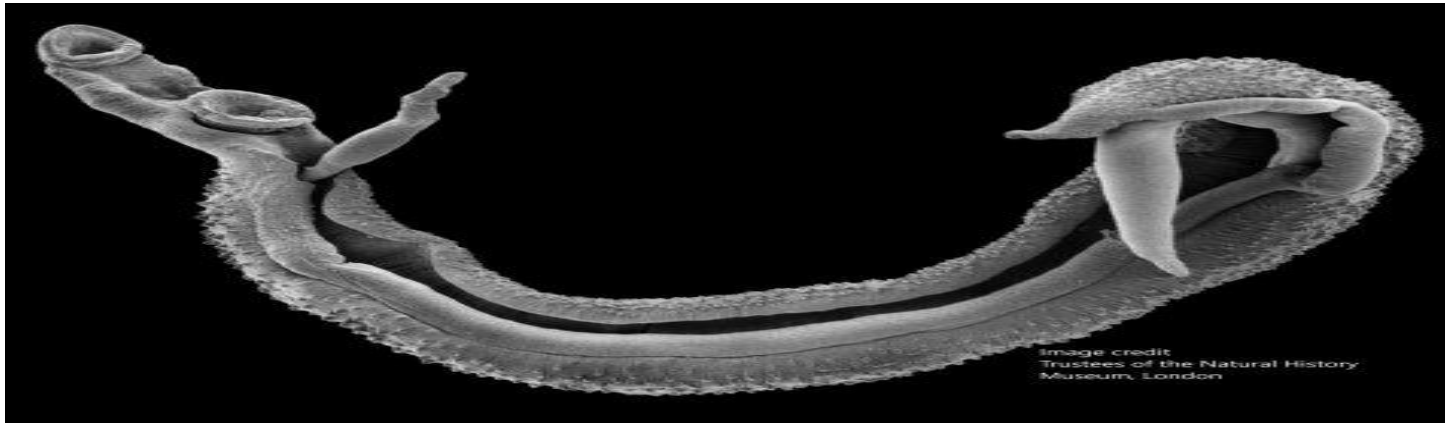
Egg: Ellipsoidal with terminal spine.

Diagnostic stage

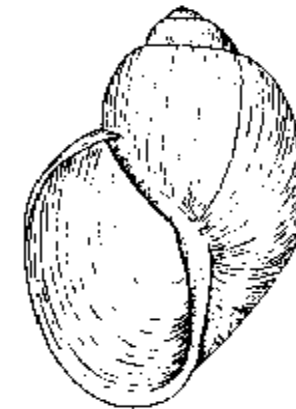


Intermediate hosts: are fresh water snails **Bulinus**

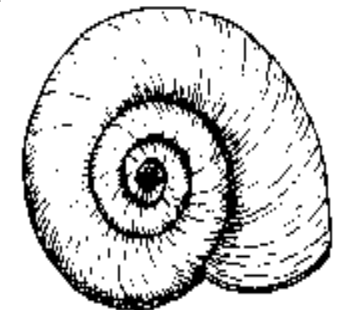




Oncomelania



Bulinus



Biomphalaria

Introduction (Cont)

- The different species of *Schistosoma* have **different types of snails** serving as their **intermediate hosts**
- Biomphalaria** for *S. mansoni* [Wikipedia](#)
- Oncomelania** for *S. japonicum* [Wikipedia](#)
- Bulinus** for *S. haematobium* [Academic Journal](#)






Galo K. Widler. Epidemiology, pathogenesis, and clinical features of schistosomiasis. *Clinical*. April 24, 2009, 1-9.

Pediatric Schistosomiasis
 Prof. Dr. Saad S.A. Ali

16/10/2018 #14

Morphology of the eggs of the schistosomes that infect humans



Schistosoma mansoni egg



Schistosoma haematobium egg



Schistosoma japonicum egg

Fasciola hepatica (liver fluke)

- Largest and commonest liver fluke.
- Large leaf-shaped with a dorsoventrally flattened body.
- Hermaphroditic parasite.
- Eggs are ovoid, and operculated.
- Definitive host: Primary definitive host is **sheep**, but it is also found in **biliary tract of man**.
- First Intermediate host: **Fresh water snails (Lymnaeae)**.
- Second Intermediate host: **Aquatic vegetations**.



Fasciola hepatica

- Infective form: **Metacercariae** encysted on raw aquatic vegetations.
- Diagnosis: Detection of eggs in stool and aspirated bile, CT, and ELISA .
- Prophylaxis: Preventing pollution of water with feces, not eating semi-aquatic plant, and not drinking fresh water from open waters.

Fasciolopsis buski (Intestinal fluke)

- *Fasciolopsis buski* Common name: Giant intestinal fluke



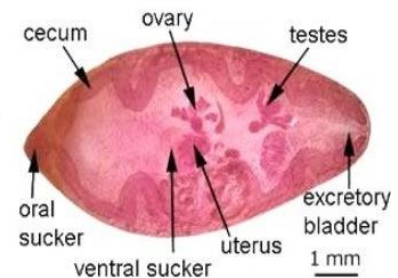
- The adult is a large fleshy worm, 20–75 mm.
- Definitive hosts: Humans, mammals
- First intermediate host: Fresh water snail, Second intermediate host: Fish
- Diagnosis is confirmed by demonstration of the egg in feces.

Lung fluke (*Paragonimus westermani*)

- Adult worm is egg-shaped, reddish, brown and covered with scale like spine.
- Habitat: Cystic spaces in the lung.
- Definitive host: Man. Besides humans, other definitive hosts include cats, tigers, foxes, dogs, pigs, beavers, and many other crab-eating mammals and domestic animals.
- First intermediate host: **Fresh water snail**
- Second intermediate host: **Fresh water crab or cray fish.**
- Infective form: **Metacercariae** encysted in crab or cray fish.

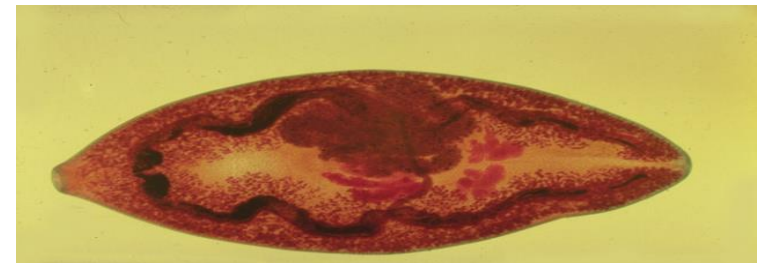
LUNG FLUKE: *PARAGONIMUS WESTERMANI*

- **Disease:** paragonimiasis
- **Distribution:** Far East, Central America, Africa, and India.
- **Morphology:** an egg-like form of the body, from 7,5 to 16 mm.



Lung fluke (*Paragonimus westermani*)

- The adult worms live in the respiratory tract of the definitive host.
- Demonstration of the eggs in sputum or feces provides definitive evidence.
- X-ray and CT scan of chest, and ELISA.
- Prevention: Adequate cooking of crabs and crayfish and washing the hands after preparing them for food.



Fasciola hepatica

intermediate host : (Snail) **lymnae**
Second Intermediate host: **Aquatic vegetations**

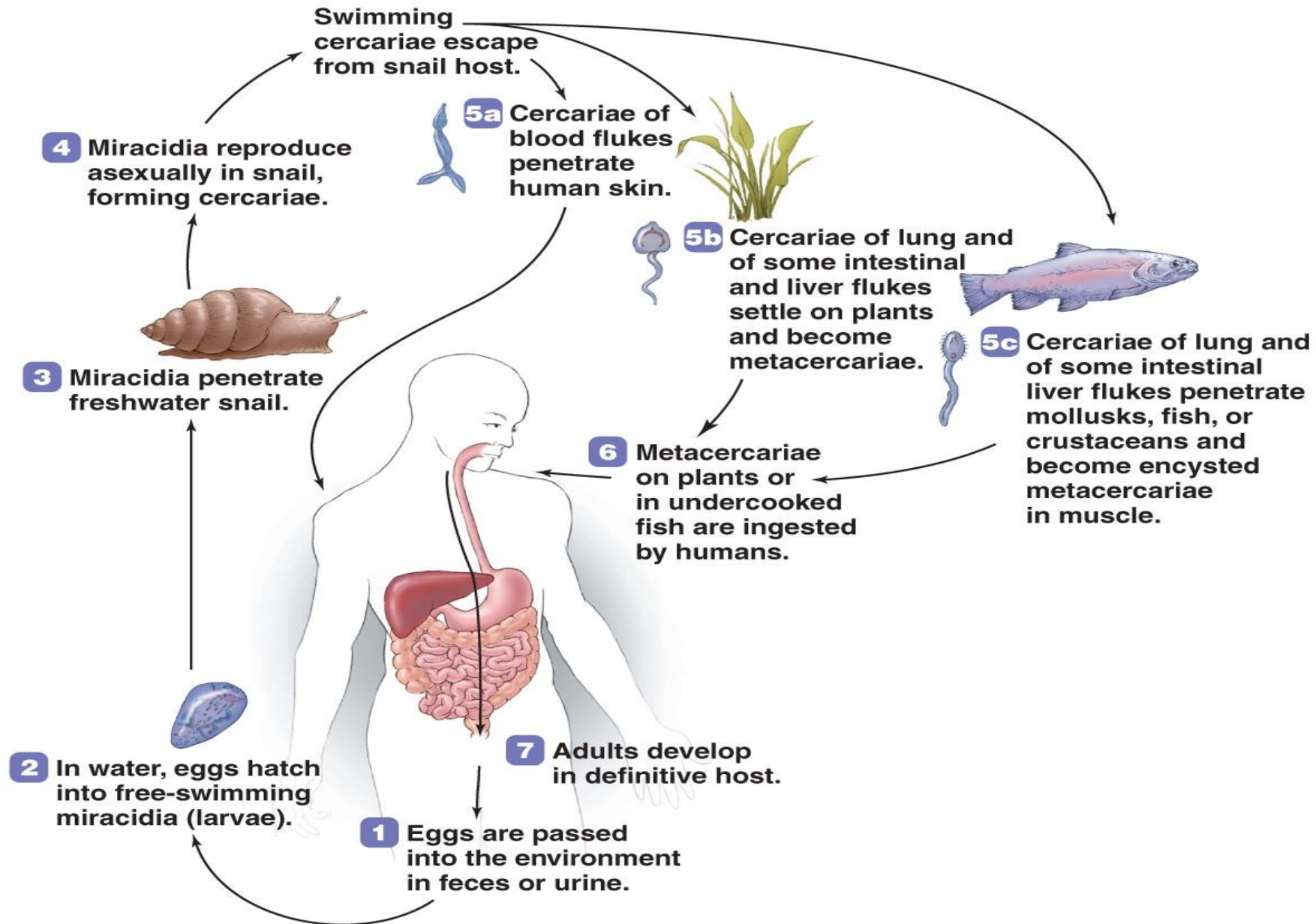


Paragonimus westermani

First intermediate host: **snails**
Second intermediate host: **crabs**



Flukes life cycles



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